Empirical Formulae & Formulae of Ionic Compounds

Exam code: 0620 & 0971

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Updated on 30 September 2025

Empirical formulae

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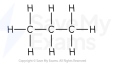
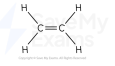
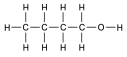
- The **empirical formula** is the simplest whole number ratio of the atoms of each element present in one molecule or formula unit of the compound
- The empirical formula of an organic molecule is often different to its molecular / chemical formula
  - For example, ethanoic acid has the chemical formula CH_3COOH or $C_2H_4O_2$ but its empirical formula is CH_2O
- The molecular / chemical formula of an ionic compound is always its empirical formula
  - For example, sodium chloride has the chemical formula NaCl, which is also its empirical formula

Worked Example

Complete the table to give the molecular and empirical formulae of the given compounds.

Chemical	Molecular formula	Empirical formula
		
		
		

Answers:

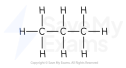
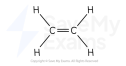
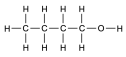
The completed table is:

Help me understand

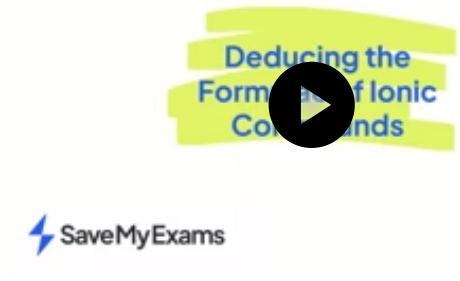
Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

	C_3H_8	C_3H_8
	C_2H_4	CH_2
	$C_4H_{10}O$	$C_4H_{10}O$

- The first compound contains 3 carbon atoms and 8 hydrogen atoms
  - This 3:8 ratio of atoms cannot be simplified
  - Therefore, the molecular and empirical formula are both C_3H_8
- The second compound contains 2 carbon atoms and 4 hydrogen atoms
  - This 2:4 ratio of atoms can be simplified to 1:2
  - Therefore, the molecular formula is C_2H_4 and the empirical formula is CH_2
- The third compound contains 4 carbon atoms, 10 hydrogen atoms and 1 oxygen atom
  - This 10:4:1 ratio of atoms cannot be simplified
  - Therefore, the molecular and empirical formula are both $C_4H_{10}O$



Did this video help you? Yes No

Deducing formulae of ionic compounds

Extended tier only

- Metals and non-metals react together to form ionic compounds
  - Ionic compounds are **not** simple molecules
  - **Remember:** Simple molecules are formed when non-metals react together to form compounds
- Ionic compounds involve the metal losing electrons and the non-metal gaining electrons to form ions

- Hydrogen ions, H^+ - sometimes referred to as protons
- Group 1 ions, e.g. Li^+, Na^+, K^+
- Group 7 ions, F^-, Cl^-, Br^-
- Copper(II) ions, Cu^{2+}
- Iron(II) ions, Fe^{2+}
- Iron(III) ions, Fe^{3+}

- There are some polyatomic (containing more than one atom) ions stated in the exam specification:
  - Carbonate ions, CO_3^{2-}
  - Sulfate ions, SO_4^{2-}
  - Hydroxide ions, OH^-
  - Nitrate ions, NO_3^-
  - Ammonium ions, NH_4^+

How to determine the formulae of ionic compounds

- Ionic compounds typically have **no overall charge**
  - This means that the size of any positively charged ion is cancelled by the size of any negatively charged ion
  - **Careful:** This should not be confused with an atom having no overall charge

Direct comparison

- The formula of an ionic compound can be determined by directly comparing the charges of the ions:
  - For example, iron(II) sulfate
    - The iron(II) ion is Fe^{2+}, which means that it has a 2+ or +2 charge
    - The sulfate ion is SO_4^{2-}, which means that it has a 2− or −2 charge
    - The charges cancel each other out Mathematically, (+2) + (−2) = 0
    - This means that one SO_4^{2-} ion is needed to cancel the +2 charge on Fe^{2+}
    - Therefore, the formula of iron(II) sulfate is $FeSO_4$

The swap-and-drop method

- When the ions in the ionic compound have **different charges**, it can be easier to use the **swap-and-drop** method
  - **Careful:** If you use this method with ions that have the same charge, then you must give the simplest whole number ratio to get the correct answer

- For example, copper(II) chloride:
  - The copper(II) ion is Cu^{2+}, which means that it has a 2+ or +2 charge
  - The chloride ion is Cl^-, which means that it has a 1− or −1 charge
  - The size of the charge on the copper(II) ion indicates the number of chloride ions needed, and the size of the charge on the chloride ion indicates the number of copper(II) ions needed

Determining the formula of copper(II) chloride

The charges swap from element to element and drop down. The positive and negative signs are removed and there is no need for the number 1.

- This gives the overall formula of copper(II) chloride as $CuCl_2$

Worked Example

The compound produced in the reaction between iron wool and chlorine contains the ions Fe^{3+} and Cl^-.

a) Give the formula of this compound.

b) State the name of this compound.

Answers:

Part a)

- **Direct comparison method:**
  - The iron ion is Fe^{3+}, which means that it has a 3+ or +3 charge
  - The chloride ion is Cl^-, which means that it has a 1− or −1 charge
  - The charges do not cancel each other out
    - Mathematically, $(+3) + (-1) \neq 0$
  - Three Cl^- ions are needed to cancel the +3 charge on Fe^{3+}
  - Therefore, the formula is **$FeCl_3$**
- **Swap-and-drop method**

- The formula is **$FeCl_3$**

Part b)

- The metal is iron and the chlorine will change to chloride
- Therefore the name is **iron chloride**

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